

Program began accepting requests in November 2013, individuals have submitted over 1,800 entries from 700 universities worldwide. Intel is also looking forward to integrate the use of boards into the curriculum of schools with some already integrating them in their 2014 syllabus. This will help the maker community in universities to develop creative projects as well as help in validation of experimental results. Among the projects noted by Intel are:

- Creating software for hardware like Google Glass and smart watches
- Monitoring health of elderly
- Brain imaging
- Stratospherical balloon project
- Lunabotics mining
- Computer vision
- Virtual reality

The most common reason for requesting Intel Galileo boards is for a variety of embedded computing applications – whether as an aide in teaching introductory Embedded Computing or Microcontroller Systems courses, or as a project platform for senior level classes. Robotics, sensor networks and controlled experiments were among the most popular proposed embedded areas. More projects can be found at <http://dgit.in/inteldemoprojects>.

To enable greater collaboration among individuals and groups, Intel has a forum at <http://dgit.in/IntelCommunityMakers> where people can interact, find solutions to problems and come up with creative ideas. Intel imagines a world where the 'digital integrates with the physical' through wearable technology.

GALILEO FOR UNIVERSITIES

To increase innovation across the entire computing spectrum, Intel will be providing 50,000 Intel® Architecture (IA) Arduino boards featuring the new Intel® Quark technology to many universities. The Intel-based microcontroller boards, in the Arduino form factor that have been the de facto standard for the maker community, will enable university students to innovate creations that will be compatible with other Intel Architecture devices in the Internet of Things ecosystem.

The program includes the following available to the university at no cost:

- Intel® Galileo boards – the first IA-based Arduino boards
- An Integrated Development Environment (IDE) for Arduino on IA
- Getting Started with Intel Galileo programming guide (Participating universities receive one free download for every five Galileo boards; additional copies may be ordered from Amazon at <http://dgit.in/IntelGalileoMatt>)
- Access to the Arduino on IA support community. Additional technical support, if needed, will be

```
int led = 13;
```

•set pin number where LED is connected using the label 'led'

```
pinMode(led, OUTPUT);
```

•set the pin 13 where LED is connected as output pin

```
digitalWrite(led, HIGH);
```

•turn ON the LED by writing logic 'HIGH' to led pin

```
delay(1000);
```

•wait for 1000 millisecond (i.e 1 second)

```
digitalWrite(led, LOW);
```

•turn OFF the LED by writing logic 'HIGH' to led pin

```
delay(1000);
```

•wait for 1000 millisecond (i.e 1 second)

•proceed to first statement of loop function i.e digitalWrite(led, HIGH);

Code explanation for Blinking of LED at Pin 13

```
char output[3];
```

•declare variable with label 'output' that can hold three alphabets

```
Serial.begin(115200);
```

•initialize serial communication at 115200 bits per second

```
system("echo '#!/usr/bin/python' > myScript.py");
```

```
system("echo 'import time' >> myScript.py");
```

```
system("echo 'for i in range(10):' >> myScript.py");
```

```
system(' echo ' with open(\' log.txt\', \' w\') as fh: >> myScript.py');
```

```
system("echo ' fh.write(\'{0}\'.format(i))' >> myScript.py");
```

```
system("echo ' time.sleep(1)' >> myScript.py");
```

•write python script myScript.py to write number 0-9 to file log.txt

```
system("chmod a+x myScript.py");
```

•make the script myScript.py executable

```
system("./myScript.py &");
```

•Execute the python script myScript.py

```
FILE *fp;
```

```
fp = fopen("log.txt", "r");
```

```
fgets(output, 2, fp);
```

```
fclose(fp);
```

•open file log.txt read the number count, write to variable output and close file

```
Serial.println(output);
```

•send the number via serial port

```
delay(1000);
```

•wait for 1 second and proceed to file opening again

Code explanation for executing Linux code on Galileo board